

DPO as Bradley–Terry MLE: A First-Principles Bridge Between Stochastic Control and Structural Estimation

Pedagogical companion notes for first-year PhD economics

(Complement to the attached report; self-contained lecture notes)

What this document is for

These notes are designed to complement the attached report by building the equivalence

$$\text{DPO policy estimation} \iff \text{Bradley–Terry MLE}$$

from first principles, with special attention to *what objects are being optimized or estimated*, what assumptions are required for the equivalence to hold, and how the same algebra reappears in recursive methods (soft Bellman equations) and structural dynamic discrete choice (Hotz–Miller inversions).

Main takeaway (in one sentence). If (i) preferences are generated by a Bradley–Terry random utility model in *reward differences* and (ii) the policy that generated those rewards is the optimizer of a *KL-regularized* control problem around a reference policy, then the *likelihood* of observed preference labels can be written purely in terms of *policy log-probability ratios*, making DPO exactly the maximum likelihood estimator for a well-defined paired-comparison model.

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1 Objects and ontological hygiene: policy vs. estimator

A lot of confusion in this literature comes from silently switching between *mathematical objects*: functions, correspondences, probability measures, and data-to-parameter maps. Let’s pin them down.

1.1 Policies are probability kernels (not parameters)

Fix a measurable context space $\mathcal{X}$ (prompts, states, environments). For each $x \in \mathcal{X}$, let $\mathcal{Y}(x)$ be the set of feasible actions/outcomes (candidate completions, action sequences, etc.). A **policy** is a conditional distribution over $\mathcal{Y}(x)$:

$$\pi(\cdot \mid x) \in \Delta(\mathcal{Y}(x)).$$

Formally, π is a Markov kernel: for each x , a probability measure on $\mathcal{Y}(x)$, and for each measurable $A \subseteq \mathcal{Y}(x)$, the map $x \mapsto \pi(A \mid x)$ is measurable.

1.2 Estimators map data to parameters (and thereby to policies)

An **econometric estimator** is a map from a dataset D to an element of a parameter space Θ ,

$$\hat{\theta} : D \rightarrow \Theta.$$

If a policy is parameterized by θ —written π_θ —then composing gives an *estimated policy* $\hat{\pi}(\cdot \mid x) := \pi_{\hat{\theta}(D)}(\cdot \mid x)$.

Remark 1 (Answering the first-principles question). An *optimal policy* and an *estimator* are not the same kind of object. A policy is a conditional distribution; an estimator is a data-dependent rule returning an element of a parameter space (or a function/correspondence). They become intimately related when we (a) posit a parametric family $\{\pi_\theta\}$ and (b) choose $\hat{\theta}$ by solving an optimization problem. Then the *output* of the estimator is a policy—but the estimator itself is still a data-to-parameter map.

Definition 1 (Policy Function). A **policy function** $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ maps states to probability distributions over actions. It is a *decision rule* that characterizes optimal behavior given primitives (u, β, δ) .

Definition 2 (Structural Estimator). A **structural estimator** $\hat{\theta}$ is a statistic computed from data $\mathcal{D}$ that recovers parameters of a behavioral model (e.g., preferences, technology).

When are they the same?

By the principle of *revealed preference*, observed choices *reveal* the underlying policy. If we observe (s_i, a_i) pairs and posit a parametric policy $\pi_\theta(a|s)$, then:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \sum_{i=1}^N \log \pi_\theta(a_i | s_i)$$

is both:

- An **estimator** of the policy parameters θ
- A characterization of the **optimal policy** under the assumption that observed behavior is optimal

Economic Intuition

The Bridge: Structural econometrics *assumes* optimality and uses data to recover the policy. Dynamic programming *derives* the policy from primitives. DPO shows these are inverses of each other: the policy that explains preference data under BT is structurally equivalent to the policy that maximizes KL-regularized expected reward.

1.3 Two “problems” whose solutions we’ll meet

We will study two models that live on the same (x, y) space but tell different stories:

- A **positive (statistical)** model: how preference labels are generated from latent utilities (Bradley–Terry / random utility).
- A **normative (control)** model: which policy would be chosen if an agent maximized reward subject to an information/control cost (KL-regularized planner).

The equivalence arises because the KL-regularized optimizer has a *log-ratio inversion* that turns reward differences into policy log-probability ratio differences. That makes the statistical likelihood depend only on the policy.

1.4 What is “Structural” About DPO?

A method is **structural** if it:

1. Posits economic primitives (preferences, technology, information)
2. Derives testable implications (choice probabilities, moments)

3. Recovers primitives from data via identification arguments

DPO is structural because:

- **Primitives:** Latent reward function $r^*(x, y)$, reference policy π_{ref} , rationality parameter β
- **Implications:** Preference probabilities $P(y_w \succ y_\ell | x) = \sigma(\beta^{-1}(r^*(x, y_w) - r^*(x, y_\ell)))$
- **Identification:** Log-ratio inversion recovers reward *differences* from observed preferences

Contrast with **reduced-form** methods like supervised fine-tuning, which directly fit $\pi_\theta(y|x)$ to observed (x, y) pairs without a behavioral model.

2 The two model cards (fixed structure, side-by-side comparability)

To prevent “symbol drift,” we use a common model-card template. Each card answers: purpose, primitives, data, assumptions, estimand, identification, computation, and failure modes.

Common notation

- $x \in \mathcal{X}$: context (prompt / state).
- $y \in \mathcal{Y}(x)$: candidate completion / action / outcome.
- $D = \{(x_i, y_i^{(w)}, y_i^{(\ell)})\}_{i=1}^N$: preference comparisons where $y^{(w)}$ is chosen over $y^{(\ell)}$ given x .
- $\pi_{\text{ref}}(\cdot \mid x)$: reference policy (baseline distribution).
- $\beta > 0$: scale/temperature parameter.

2.1 Model card A: Bradley–Terry random utility model (paired comparisons)

Model card A: Bradley–Terry (BT) / logistic paired comparison

Purpose. A tractable probabilistic model for noisy pairwise preference labels.

Primitives.

- A real-valued score (latent utility / reward) $r^* : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$.
- Logistic link $\sigma(u) := (1 + e^{-u})^{-1}$.

Data. Observations $\{(x_i, y_i^{(w)}, y_i^{(\ell)})\}_{i=1}^N$ where $y_i^{(w)}$ is labeled preferred.

Model (likelihood).

$$\Pr(y^{(w)} \succ y^{(\ell)} \mid x) = \sigma(r^*(x, y^{(w)}) - r^*(x, y^{(\ell)})).$$

Estimand. Utility *differences* (or parameters indexing them). Full levels of $r^*(x, \cdot)$ are not identified.

Identification / invariances.

- (Additive) For each fixed x , $r^*(x, \cdot)$ is identified only up to $r^*(x, y) \mapsto r^*(x, y) + c(x)$.
- (Scale) If β is introduced as a separate preference-noise scale, only the ratio r^*/β matters absent normalization.

Computation. Convex (in linear scores) logistic regression on differences; nonconvex if r^* is neural/LLM-parameterized.

Failure modes. Context effects, non-transitivity, rater heterogeneity, dependence on the *presented* choice set, or systematic violations of the logit error structure imply misspecification. The MLE remains an M -estimator for the best-fitting logit model, but “utility” becomes an approximation.

Microfoundation: Random Utility Maximization

The Bradley-Terry model has a clean microfoundation in random utility theory:

Theorem 1 (Luce-McFadden). *Assume agent i has latent utility:*

$$U_i(x, y) = r^*(x, y) + \varepsilon_y$$

where $\{\varepsilon_y\}_{y \in \mathcal{Y}(x)}$ are i.i.d. Type-I Extreme Value (Gumbel) random variables with CDF $F(\varepsilon) = \exp(-\exp(-\varepsilon))$.

Then:

$$P(y_1 \succ y_2 | x) = P(U(x, y_1) \geq U(x, y_2)) = \sigma(r^*(x, y_1) - r^*(x, y_2))$$

Proof Sketch.

$$\begin{aligned} P(y_1 \succ y_2) &= P(r^*(x, y_1) + \varepsilon_1 \geq r^*(x, y_2) + \varepsilon_2) \\ &= P(\varepsilon_2 - \varepsilon_1 \leq r^*(x, y_1) - r^*(x, y_2)) \end{aligned}$$

The difference of two i.i.d. Gumbel random variables has a logistic distribution:

$$P(\varepsilon_2 - \varepsilon_1 \leq z) = \frac{e^z}{1 + e^z} = \sigma(z)$$

□

Maximum Likelihood Estimation

Data: $\mathcal{D} = \{(x_i, y_i^{(w)}, y_i^{(\ell)})\}_{i=1}^N$ where $y^{(w)}$ was preferred to $y^{(\ell)}$ given context x .

Parametric Model: Posit $r^*(x, y) = r_\theta(x, y)$ for some parameter vector θ .

Log-Likelihood:

$$\ell(\theta; \mathcal{D}) = \sum_{i=1}^N \log \sigma(r_\theta(x_i, y_i^{(w)}) - r_\theta(x_i, y_i^{(\ell)}))$$

MLE:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta; \mathcal{D})$$

Identification

Proposition 1 (Identification Up to Location). *For each context x , the reward function $r^*(x, \cdot)$ is identified only up to an additive constant $c(x)$, since preference probabilities depend only on differences:*

$$\sigma(r^*(x, y_1) - r^*(x, y_2)) = \sigma((r^*(x, y_1) + c(x)) - (r^*(x, y_2) + c(x)))$$

Implication: We can never recover absolute rewards from preference data alone—only reward *rankings* within each context.

Economic Interpretation

- $r^*(x, y)$: Systematic utility component
- ε_y : Unobserved heterogeneity, measurement error, or “trembles”
- **Gumbel assumption:** IIA (Independence of Irrelevant Alternatives)—adding a third option doesn’t change relative probabilities of first two

2.2 Model card B: KL-regularized planner and the tilted Gibbs (softmax) policy

Model card B: KL-regularized (entropy) control around a reference policy

Purpose. A normative model of policy choice trading off expected reward and deviation from a baseline distribution (information cost / control cost / maximum entropy regularization).

Primitives.

- A reward (or utility) function $r^*(x, y)$.
- A reference policy $\pi_{\text{ref}}(\cdot | x)$ with $\pi_{\text{ref}}(y | x) > 0$ on the relevant support.
- A regularization parameter $\beta > 0$ governing the reward–KL tradeoff.

Problem (pointwise in x). Choose $\pi(\cdot | x)$ to solve

$$\max_{\pi(\cdot | x)} \mathbb{E}_{y \sim \pi(\cdot | x)}[r^*(x, y)] - \beta \text{KL}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) \quad \text{s.t.} \quad \sum_y \pi(y | x) = 1. \quad (1)$$

(Replace sums by integrals if $\mathcal{Y}(x)$ is continuous; the implemented estimand is typically finite-candidate.)

Solution (tilted Gibbs / softmax). If $r^*(x, \cdot)$ is bounded (or integrable) and π_{ref} has full support, the objective is strictly concave and the unique maximizer satisfies

$$\pi^*(y | x) = \frac{\pi_{\text{ref}}(y | x) \exp(r^*(x, y)/\beta)}{Z(x)}, \quad Z(x) = \sum_{y'} \pi_{\text{ref}}(y' | x) \exp(r^*(x, y')/\beta). \quad (2)$$

Key inversion (log-ratio identity). Taking logs of (2) yields, for any y_1, y_2 ,

$$r^*(x, y_1) - r^*(x, y_2) = \beta \left[\log \frac{\pi^*(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \log \frac{\pi^*(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)} \right]. \quad (3)$$

The unknown normalizer $\beta \log Z(x)$ cancels in differences.

Interpretation. π_{ref} acts like a baseline “availability” (or prior). The KL term penalizes information/control used to steer the policy away from the baseline. β plays the role of a shock scale / rational inattention parameter.

Failure modes. If $\pi_{\text{ref}}(y | x) = 0$ on outcomes that matter, log-ratios diverge and the model breaks. If the control cost is not KL (e.g., other divergences), the closed-form (2) and inversion (3) change.

The Control Problem

Definition 3 (RLHF-KL Objective). Given a reference policy $\pi_{\text{ref}} : \mathcal{X} \rightarrow \Delta(\mathcal{Y})$ and a reward function $r^* : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, the KL-regularized optimization problem is:

$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot | x)}[r^*(x, y)] - \beta \cdot D_{\text{KL}}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))$$

where the KL divergence is:

$$D_{\text{KL}}(\pi \parallel \pi_{\text{ref}}) = \sum_{y \in \mathcal{Y}(x)} \pi(y|x) \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)}$$

Interpretation

Why KL regularization?

1. **Robustness:** Prevent overfitting to potentially noisy human preferences
2. **Fluency:** Stay close to the pre-trained language model distribution
3. **Computational:** Avoid extreme probability ratios that cause training instability
4. **Rational Inattention:** β^{-1} parameterizes the cost of deviating from the reference

Economic Intuition

Think of π_{ref} as a “default” or “baseline” behavioral pattern (e.g., habitual consumption). The agent optimizes rewards but faces a psychological or computational cost (measured by KL divergence) of deviating from this default. As $\beta \rightarrow 0$, the agent becomes infinitely rational (ignores the reference). As $\beta \rightarrow \infty$, the agent is boundedly rational (sticks to the reference).

Analytical Solution: The Tilted Gibbs Distribution

Theorem 2 (Closed-Form Optimal Policy). *The unique optimal policy for the RLHF-KL problem is:*

$$\pi^*(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{r^*(x, y)}{\beta}\right)$$

where $Z(x) = \sum_{y'} \pi_{\text{ref}}(y'|x) \exp(r^*(x, y')/\beta)$ is the partition function.

Proof. Form the Lagrangian:

$$\mathcal{L}(\pi, \lambda) = \sum_y \pi(y|x) r^*(x, y) - \beta \sum_y \pi(y|x) \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} + \lambda \left(\sum_y \pi(y|x) - 1 \right)$$

First-order condition w.r.t. $\pi(y|x)$:

$$0 = r^*(x, y) - \beta \left(\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} + 1 \right) + \lambda$$

Rearrange:

$$\log \pi(y|x) = \frac{r^*(x, y)}{\beta} + \log \pi_{\text{ref}}(y|x) + \frac{\lambda - \beta}{\beta}$$

Exponentiate and normalize using $\sum_y \pi(y|x) = 1$ to get the stated form. $\square$

The Log-Ratio Inversion (Key Step)

Corollary 3 (Reward Differences as Log-Ratios). *Taking logs of the optimal policy:*

$$\log \pi^*(y|x) = \frac{r^*(x,y)}{\beta} + \log \pi_{ref}(y|x) - \log Z(x)$$

Therefore, for any two alternatives y_1, y_2 :

$$\begin{aligned} r^*(x, y_1) - r^*(x, y_2) &= \beta [\log \pi^*(y_1|x) - \log \pi_{ref}(y_1|x)] \\ &\quad - \beta [\log \pi^*(y_2|x) - \log \pi_{ref}(y_2|x)] \end{aligned}$$

The partition function cancels in differences!

Key Point

This is the **inversion formula**: reward differences can be expressed as log-ratios of the optimal policy relative to the reference, without needing to know $Z(x)$.

3 The equivalence: assumptions, necessity/sufficiency, and the algebra

We now isolate the exact conditions under which “DPO is Bradley–Terry MLE” is literally true.

3.1 Assumptions (minimal and explicit)

- (A1) **BT data-generating process.** Conditional on (x, y_1, y_2) , the probability that a labeler prefers y_1 to y_2 is $\sigma(r^*(x, y_1) - r^*(x, y_2))$ for some latent r^* .
- (A2) **KL-optimality relation.** The policy that “implements” those rewards is the optimizer of the KL-regularized planner (1) with reference π_{ref} and temperature β , giving (2).
- (A3) **Support compatibility.** $\pi_{\text{ref}}(y | x) > 0$ wherever the policy family (and the data) place mass, so that log-ratios are well-defined.
- (A4) **Estimation family.** We fit a parametric policy $\pi_\theta(\cdot | x)$ by maximum likelihood under the implied model. (Correct specification $\exists \theta^* : \pi_{\theta^*} = \pi^*$ is *not* needed for objective equivalence, only for “recover the true policy” statements.)

3.2 Proposition: DPO objective equals BT log-likelihood after substitution

Proposition 2 (DPO $\equiv$ BT MLE for log-ratio scores). *Under (A1)–(A4), the BT likelihood for preference data can be written as a function of policy log-ratios:*

$$\Pr(y^{(w)} \succ y^{(\ell)} | x) = \sigma\left(\beta \left[\log \frac{\pi^*(y^{(w)} | x)}{\pi_{\text{ref}}(y^{(w)} | x)} - \log \frac{\pi^*(y^{(\ell)} | x)}{\pi_{\text{ref}}(y^{(\ell)} | x)} \right]\right).$$

Replacing π^ by π_θ and maximizing the resulting log-likelihood yields exactly the DPO estimator:*

$$\hat{\theta}_{\text{DPO}} \in \arg \max_{\theta} \sum_{i=1}^N \log \sigma\left(\beta \left[\log \frac{\pi_\theta(y_i^{(w)} | x_i)}{\pi_{\text{ref}}(y_i^{(w)} | x_i)} - \log \frac{\pi_\theta(y_i^{(\ell)} | x_i)}{\pi_{\text{ref}}(y_i^{(\ell)} | x_i)} \right]\right). \quad (4)$$

Proof sketch, with the only nontrivial step highlighted. Start from the BT model (A1):

$$\Pr(y^{(w)} \succ y^{(\ell)} | x) = \sigma(r^*(x, y^{(w)}) - r^*(x, y^{(\ell)})).$$

Use (A2) to express reward differences by the log-ratio identity (3). The normalizer $\beta \log Z(x)$ cancels. Substitute the resulting expression into the BT probability and take logs across observations. The optimization problem in θ is precisely (4). The highlighted step is the inversion (3), which is a direct algebraic consequence of the KL-regularized optimality condition. $\square$

Step 1: Invert the Policy to Express Rewards

From the closed-form solution:

$$\log \pi^*(y | x) = \frac{r^*(x, y)}{\beta} + \log \pi_{\text{ref}}(y | x) - \log Z(x)$$

Rearrange to solve for $r^*(x, y)$:

$$r^*(x, y) = \beta \log \frac{\pi^*(y | x)}{\pi_{\text{ref}}(y | x)} + \beta \log Z(x)$$

Step 2: Compute Reward Differences

For any pair (y_w, y_ℓ) :

$$\begin{aligned} r^*(x, y_w) - r^*(x, y_\ell) &= \beta \log \frac{\pi^*(y_w|x)}{\pi_{\text{ref}}(y_w|x)} + \beta \log Z(x) \\ &\quad - \left[\beta \log \frac{\pi^*(y_\ell|x)}{\pi_{\text{ref}}(y_\ell|x)} + \beta \log Z(x) \right] \\ &= \beta \left[\log \frac{\pi^*(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi^*(y_\ell|x)}{\pi_{\text{ref}}(y_\ell|x)} \right] \end{aligned}$$

Observation: The partition function $Z(x)$ cancels! This is *crucial*—we don’t need to compute the intractable normalization constant.

Step 3: Substitute into the Bradley-Terry Likelihood

The preference probability becomes:

$$\begin{aligned} P(y_w \succ y_\ell | x) &= \sigma(r^*(x, y_w) - r^*(x, y_\ell)) \\ &= \sigma \left(\beta \left[\log \frac{\pi^*(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi^*(y_\ell|x)}{\pi_{\text{ref}}(y_\ell|x)} \right] \right) \end{aligned}$$

Step 4: Write the Parametric Log-Likelihood

We don’t observe π^* , so we posit a parametric family π_θ and maximize:

$$\ell(\theta; \mathcal{D}) = \sum_{i=1}^N \log \sigma \left(\beta \left[\log \frac{\pi_\theta(y_i^{(w)}|x_i)}{\pi_{\text{ref}}(y_i^{(w)}|x_i)} - \log \frac{\pi_\theta(y_i^{(\ell)}|x_i)}{\pi_{\text{ref}}(y_i^{(\ell)}|x_i)} \right] \right)$$

This is exactly the DPO loss function.

Step 5: What Are We Estimating?

- We are *not* estimating r^* directly
- We are estimating the policy π_θ whose log-ratio scores best explain the preference data under the BT model
- Under the structural assumption that this policy satisfies KL-regularized optimality, the estimated policy *implicitly* encodes the reward function via the log-ratio identity

Key Point

The Isomorphism:

MLE for π_θ under BT preferences $\Leftrightarrow$ Fitting the optimal policy under RLHF-KL

These are the same optimization problem. DPO *is* structural estimation.

4 A one-page comparison: BT MLE vs. KL-regularized policy (and the isomorphism)

The goal here is to let your brain see the mapping without doing any algebra.

Bradley–Terry (BT) MLE	KL-regularized tilted-Gibbs policy
Observable input Preference labels $(x, y^{(w)}, y^{(\ell)})$	Observable input Reference policy $\pi_{\text{ref}}(\cdot x)$ and temperature β
Latent object Score/utility $r^*(x, y)$ entering choice probabilities	Latent object Reward $r^*(x, y)$ entering planner’s objective
Core assumption $\Pr(y_1 \succ y_2 x) = \sigma(r^*(x, y_1) - r^*(x, y_2))$	Core assumption $\pi^*(\cdot x) \in \arg \max_{\pi} \mathbb{E}_{\pi}[r^*] - \beta \text{KL}(\pi \pi_{\text{ref}})$
Optimization problem Maximize log-likelihood over parameters for r_{η} or π_{θ}	Optimization problem Maximize regularized expected reward over π
Characterizing solution FOC gives logistic regression in score differences (convex if linear)	Characterizing solution FOC gives softmax around reference: $\pi^* \propto \pi_{\text{ref}} \exp(r^*/\beta)$
Computing solution Gradient ascent on $\sum \log \sigma(\Delta r)$	Computing solution Closed form for π^* given r^* ; iterative methods only if r^* is implicit
Interpreting parameters Utilities identified up to additive constants (and scale)	Interpreting parameters Policy trades reward vs. information/control cost; β controls deviation from π_{ref}
Isomorphism (the bridge) Substitute Δr^* with policy log-ratio difference	Isomorphism (the bridge) Invert the softmax: $\Delta r^* = \beta \Delta \log \frac{\pi^*}{\pi_{\text{ref}}}$
Resulting estimand MLE over policy parameters θ using <i>only</i> $\log(\pi_{\theta}/\pi_{\text{ref}})$	Resulting estimand Optimal policy for r^* can be expressed without r^* levels (only differences matter)

5 So what is actually being “estimated” in DPO?

There are three closely related but distinct answers. Confusing them causes most of the conceptual bugs.

5.1 (1) Policy estimation: the direct object

DPO is an M -estimator for the policy parameters θ in π_{θ} under a paired-comparison likelihood. That is the literal statistical statement: you pick θ to maximize (4).

5.2 (2) Utility differences: an implied object

Given an estimated policy $\hat{\pi}$, you can *define* an implied reward (up to constants)

$$\hat{r}(x, y) := \beta \log \frac{\hat{\pi}(y | x)}{\pi_{\text{ref}}(y | x)} + \beta \hat{c}(x),$$

where $\hat{c}(x)$ absorbs the unknown $\log Z(x)$. This recovers only differences in r , which is all BT identifies.

5.3 (3) A structural control model: a maintained interpretation

When you treat (1) as the *structural* problem, DPO is estimating the policy that would be optimal for some reward function under a KL control cost. The control model adds interpretation: π_{ref} is a prior, β is an information cost scale, and the log-ratio plays the role of a (generalized) “value.” But interpretation is only as good as the maintained assumptions.

6 Recursive methods and structural estimation: where the same algebra reappears

In recursive macroeconomics (SLP, Ljungqvist-Sargent), we:

1. Define state space $\mathcal{S}$ and action space $\mathcal{A}$
2. Write the Bellman equation: $V(s) = \max_a [u(s, a) + \beta \mathbb{E}[V(s')]]$
3. Solve via value function iteration or policy iteration
4. Derive policy function: $\pi^*(s) = \arg \max_a [u(s, a) + \beta \mathbb{E}[V(s')]]$

The DPO framework extends this to stochastic policies with entropy regularization:

Theorem 4 (Soft Bellman Equation). *Define the soft value function:*

$$V^*(s) = \beta \log \sum_{a \in \mathcal{A}} \pi_{\text{ref}}(a|s) \exp \left(\frac{r(s, a) + \gamma \mathbb{E}[V^*(s')]}{\beta} \right)$$

The optimal policy is:

$$\pi^*(a|s) = \frac{\pi_{\text{ref}}(a|s) \exp((r(s, a) + \gamma \mathbb{E}[V^*(s')])/\beta)}{\sum_{a'} \pi_{\text{ref}}(a'|s) \exp((r(s, a') + \gamma \mathbb{E}[V^*(s')])/\beta)}$$

Intuition: The $\log \sum \exp$ is a “soft max” operator. As $\beta \rightarrow 0$, this converges to the standard Bellman equation. The reference policy π_{ref} weights the alternatives.

6.1 Why Gumbel Shocks Matter

The Type-I Extreme Value (Gumbel) distribution is special because:

$$\mathbb{E} \left[\max_a \{r(s, a) + \varepsilon_a\} \right] = \beta \log \sum_a \exp(r(s, a)/\beta) + \text{const}$$

This is why:

- The soft Bellman equation has a log-sum-exp form

- The choice probabilities are logit/softmax
- Value function iteration remains a contraction

Economic Intuition

Gumbel shocks give us *closed-form aggregation* of option values. The “inclusive value” or “log-sum-exp” term represents the expected value of the best choice under uncertainty. This same mathematics appears in:

- McFadden’s logit demand
- Rust’s dynamic discrete choice
- Soft Q-learning in RL
- DPO for language model alignment

6.2 The Hotz-Miller Connection

Hotz and Miller (1993) showed that in dynamic discrete choice models with logit shocks:

$$V(s, a) - V(s, a') = \log P(a|s) - \log P(a'|s) + \log \nu(a) - \log \nu(a')$$

where $\nu(a)$ are known functions of the shock distribution.

DPO uses the same logic:

$$r^*(x, y) - r^*(x, y') = \beta \left[\log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} - \log \frac{\pi^*(y'|x)}{\pi_{\text{ref}}(y'|x)} \right]$$

The reference policy π_{ref} plays the role of the base measure, and β is the scale parameter.

7 Diagnostics and failure modes: when the equivalence fails (and how to notice)

The equivalence is sharp. It breaks in predictable ways.

Checklist: what must be true for the algebra to go through

- **(BT structure)** Preference labels must be representable as $\sigma(\Delta r)$ for some r .
- **(KL control cost)** The planner must use KL divergence to the stated π_{ref} (not another regularizer).
- **(Shared reward)** The same r^* must appear in both the preference model and the control problem.
- **(Support)** $\pi_{\text{ref}} > 0$ on relevant outcomes so log-ratios exist.
- **(Finite candidate implementation)** In practice, the estimand is defined on sampled candidates; interpret accordingly.

Common subtlety: the candidate set matters. BT/logit is naturally a model over a fixed choice set. In LLM training, each x effectively comes with a *sampled* set of candidates used for labeling. The cleanest interpretation is: you are estimating the policy restricted to that sampled set (or the induced distribution over it), and extrapolation beyond it is model-dependent.

Rater heterogeneity. If different labelers have different reward functions r_j^* , the BT model becomes a mixture. DPO then targets a compromise policy that best fits the aggregated labels under the assumed single- r^* logistic model.

8 Worked micro-example (finite candidate set)

Fix a context x and three candidates $\mathcal{Y}(x) = \{a, b, c\}$. Suppose a reference policy is $\pi_{\text{ref}}(\cdot | x) = (0.6, 0.3, 0.1)$ and the reward levels are $r^*(x, \cdot) = (0, 1, 2)$ with $\beta = 1$. Then the optimal policy is

$$\pi^*(y | x) \propto \pi_{\text{ref}}(y | x) e^{r^*(x, y)}.$$

Compute unnormalized weights: $w_a = 0.6e^0 = 0.6$, $w_b = 0.3e^1 \approx 0.815$, $w_c = 0.1e^2 \approx 0.739$, so $\pi^* \approx (0.28, 0.38, 0.34)$ after normalization. The implied BT probability that c beats a is $\sigma(r_c - r_a) = \sigma(2) \approx 0.88$. Using log-ratios instead:

$$\beta \left(\log \frac{\pi^*(c)}{\pi_{\text{ref}}(c)} - \log \frac{\pi^*(a)}{\pi_{\text{ref}}(a)} \right) = \log \frac{0.34}{0.1} - \log \frac{0.28}{0.6} \approx \log(3.4) - \log(0.47) \approx 1.22 - (-0.76) = 1.98,$$

which matches $\Delta r = 2$ up to rounding. Substitution reproduces the same BT probability. This is the equivalence in arithmetic form.

9 Application: Synthetic Cultural Agents

9.1 The Cultural Calibration Problem

Standard LLMs are trained on WEIRD (Western, Educated, Industrialized, Rich, Democratic) data. To model non-WEIRD populations, we need to:

1. Define a cultural state vector z parameterizing preferences
2. Construct a reward function $R(x, y; z)$ that varies with culture
3. Estimate a policy $\pi(y|x, z)$ that maximizes this reward

9.2 The GPS Framework

Following Falk et al. (QJE 2018), we define:

$$z = \begin{pmatrix} \delta & \text{(time preference)} \\ \gamma & \text{(risk aversion)} \\ \xi^+ & \text{(positive reciprocity)} \\ \xi^- & \text{(negative reciprocity)} \\ \alpha & \text{(altruism)} \\ \tau & \text{(trust)} \end{pmatrix}$$

These are not ad-hoc labels but *structural parameters* estimated from incentivized experiments in 76 countries.

9.3 The Synthetic Ethnography Pipeline

Problem: For many populations, we lack digital trace data to train models directly.

Solution: Use the DPO isomorphism to create synthetic preference data:

1. **Teacher Model:** Condition GPT-5/Claude/Gemini/Grok on ethnographic profiles and GPS parameters
2. **Generate Synthetic Preferences:** Produce (x, y_w, y_ℓ) triplets reflecting cultural preferences
3. **Student Model:** Use DPO to distill these preferences into a smaller, open-source model (e.g., nemotron, mixtral, gpt-oss)
4. **Validation:** Check that the student’s behavior matches held-out moments from WVS, GPS, etc.

9.4 Why This Works

The DPO isomorphism means that:

- Synthetic preference data *encodes* the reward function $R(x, y; z)$
- The student learns the optimal policy $\pi^*(\cdot|x, z)$ without needing an explicit reward model
- Cultural parameters z shift the induced policy in a structurally interpretable way

This is *transfer learning as structural estimation*—we’re using a frontier model’s world knowledge to calibrate behavioral parameters for populations outside its training distribution.

10 Conclusion and Further Directions

10.1 What We’ve Learned

1. **Optimal policies and MLE estimators are the same object** under revealed preference logic with KL-regularized optimality.
2. **DPO is a structural method**, not an ad-hoc ML trick. It recovers policies via inversion of choice probabilities—exactly the strategy used in structural DDC.
3. **The Bradley-Terry model and KL-regularized RL are isomorphic** when rewards are expressed as log-ratios relative to a reference policy.
4. **Recursive methods and structural estimation connect** through the Hotz-Miller inversion: both avoid repeatedly solving fixed-point problems by exploiting closed-form relationships between policies and values.
5. **Cultural alignment can be formalized** using GPS-calibrated reward functions and synthetic ethnography via teacher-student distillation.

10.2 Open Questions for Further Study

1. **Misspecification:** What happens when preferences violate BT (e.g., intransitivity, context effects)? DPO remains a valid M-estimator, but what is the *economic interpretation* of the fitted policy?
2. **Dynamics:** The one-step analysis extends to sequential decision-making via soft Bellman equations. How do we estimate *recursive* cultural preferences over trajectories?
3. **Heterogeneity:** Real populations have distributions over z . Can we extend DPO to mixture models or hierarchical specifications?
4. **Counterfactuals:** Once we have $\hat{\pi}(y|x, z)$, how do we conduct policy analysis? What is the “treatment effect” of changing cultural parameters?
5. **Identification:** What variation in the data identifies different components of z ? This is a classic structural IV problem.

10.3 Suggested Readings

Structural Econometrics:

- Hotz & Miller (1993): “Conditional Choice Probabilities and the Estimation of Dynamic Models”
- Rust (1987): “Optimal Replacement of GMC Bus Engines”
- Train (2009): *Discrete Choice Methods with Simulation*

Recursive Methods:

- Stokey, Lucas & Prescott (1989): *Recursive Methods in Economic Dynamics*
- Ljungqvist & Sargent (2018): *Recursive Macroeconomic Theory*

Machine Learning & Alignment:

- Rafailov et al. (2023): “Direct Preference Optimization: Your Language Model is Secretly a Reward Model”

- Falk et al. (2018): “Global Evidence on Economic Preferences”
- Ouyang et al. (2022): “Training Language Models to Follow Instructions with Human Feedback”